SKETCH TO FACE CRIMINAL IDENTIFICATION

AND REGISTRATION SYSTEM

B. Tech. Project End Sem Report

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CERTIFICATE

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ABSTRACT

Crime investigation often relies on forensic sketches to identify suspects in the absence of direct photographic evidence. However, traditional sketches lack fine details, making identification challenging. This report presents an intelligent system that enhances suspect identification by matching forensic sketches with a structured image dataset. The system analyzes key facial features and retrieves the closest matching colored image from the dataset, bridging the gap between rough sketches and real-life appearances. By automating the matching process, the system significantly improves identification accuracy and reduces manual efforts in investigations.

Additionally, the proposed system integrates an efficient image retrieval mechanism that refines search results based on facial attributes, ensuring precise suspect identification. The approach not only accelerates case resolutions but also minimizes human bias in suspect recognition. With its ability to process vast datasets quickly, this system provides law enforcement agencies with a reliable and scalable tool for modern crime-solving. Furthermore, its application extends beyond criminal investigations to areas such as missing person identification and enhanced security monitoring. The development of such technology marks a significant advancement in forensic science, bridging the gap between forensic artistry and real-world identification systems.

Contents

Chapter 1

Introduction

1.1 Introduction

Forensic investigations often rely on sketches to identify suspects when photographic evidence is unavailable. These sketches, typically drawn based on eyewitness descriptions, serve as a critical tool in criminal identification but are limited by their lack of fine details and dependency on human memory. The advent of artificial intelligence (AI) and deep learning offers a transformative opportunity to enhance forensic sketch-based identification. This project introduces a Sketch-to-Face Criminal Identification and Registration System that leverages advanced deep learning techniques to convert forensic sketches into photorealistic facial images, thereby improving the accuracy and efficiency of suspect identification in law enforcement.

1.2 Existing System

Current forensic identification systems primarily depend on manual comparison of sketches with criminal databases or composite sketch software used by forensic artists. These systems are labor-intensive, time-consuming, and prone to errors due to inconsistencies in sketch quality and subjective interpretations. Some AI-based solutions, such as feature-based facial recognition and early neural network models, have been explored, but they struggle with the modality gap between sketches and real images, limited dataset availability, and variations in sketch styles. These shortcomings highlight the need for an automated, robust system capable of bridging the gap between forensic sketches and real-world facial images.

1.3 Problem Definition

Forensic sketch-based identification is a crucial aspect of criminal investigations, especially when security cameras fail to capture clear images or when no photographic evidence is available. Traditional forensic sketches, often created based on eyewitness accounts, lack intricate details, making suspect recognition challenging. Manually comparing these sketches with criminal databases is time-consuming and prone to errors due to significant differences between sketches and real photographs. This research addresses these challenges by employing deep learning-based techniques to enhance forensic sketch-to-image transformation.

1.4 Objective

The primary objective of this study is to develop an advanced Sketch-to-Face transformation model using deep learning techniques. The model aims to:

- Enhance forensic sketches by generating high-quality, realistic facial images.

- Reduce the modality gap between sketches and real images.
- Improve identification accuracy using CNN, ResNet-50, Siamese Neural Networks, and AlexNet.
- Automate the process of comparing sketches with criminal databases, accelerating suspect identification.

1.5 Outline

This report outlines the development and evaluation of the Sketch-to-Face Criminal Identification and Registration System across several chapters, covering literature review, research gaps, methodology, experimental setup, results, and conclusions, followed by appendices with supplementary details.

The next chapter, the Literature Review, examines prior work in forensic sketch-based identification and facial recognition. It categorizes studies into AI-based, traditional, and alternative approaches, highlighting their strengths, limitations, and the need for an advanced deep learning-based system to improve suspect identification accuracy.

Chapter 2

Literature Survey And Research Gap

2.1 Introduction

Forensic sketch-based suspect identification plays a critical role in criminal investigations where photographic evidence is lacking. Traditionally, these sketches are manually compared against criminal databases, a process that is both slow and error-prone. Recent advancements in artificial intelligence (AI), machine learning (ML), and deep learning (DL) have led to the development of automated methods aimed at improving the accuracy and efficiency of forensic sketch-to-photo matching. These techniques range from traditional manual methods to advanced deep learning-based systems.

2.2 Overview of Automated Recognition Techniques

Various methodologies have emerged to address the limitations of manual matching:

- **Traditional manual matching:** Involves visually comparing sketches with photographic databases.

- **Feature-based methods:** Utilize facial landmarks and handcrafted features for recognition.

- **Deep learning techniques:** Employ CNNs and GANs to produce photorealistic images from sketches.

2.3 Research Gap

Despite the promise of AI-based approaches, there are critical challenges that hinder their full deployment in real-world scenarios:

2.3.1 Limitations of Traditional Methods

- Sketches are manually matched, which is time-consuming and unreliable.

- Eyewitness accounts are subjective and often lack detailed features.

2.3.2 Challenges in Existing AI Methods

- A significant modality gap exists between sketches and real images.

- Datasets with paired sketch-photo images are limited in size and diversity.

- AI models often struggle with pose, lighting, and stylistic variations.

2.3.3 Need for Advanced Solutions

An effective system must:

- Minimize modality discrepancies using CNNs, GANs, ResNet-50, and Siamese Networks.
- Be adaptable to real-time applications and varied forensic scenarios.
- Continuously improve with incoming forensic data.

2.4 Outline

This research builds on the identified gaps and outlines a hybrid deep learning system for sketch-to-face transformation. The following chapters will elaborate on the methodology, experimental results, evaluation metrics, and implications for forensic science.

Chapter 3 will describe the architectural design of the hybrid system, including data preprocessing, model training, and integration strategies for CNNs, GANs, and Siamese Networks.

In this chapter, we will explore the components that enable high-quality sketch-to-image transformations, with emphasis on optimizing neural network parameters and enhancing generalization across diverse forensic datasets.

Chapter 3

Proposed Methodology

The proposed methodology aims to enhance forensic sketch-based suspect identification by transforming forensic sketches into realistic facial images using deep learning. This system integrates multiple neural networks to bridge the gap between forensic sketches and real images, improving recognition accuracy and efficiency in law enforcement.

3.1 Dataset Description

The dataset used in this study consists of forensic sketches and corresponding real images. It is sourced from:

- **Kaggle Dataset:** Contains high-resolution facial images with diverse features.
- **Custom Dataset:** Includes forensic sketches paired with real-world images for model training and evaluation.

Each image is labeled with a "Match Confidence Score", representing the similarity between the forensic sketch and real image.

3.2 Data Preprocessing

3.2.1 Data Augmentation

To improve model generalization, various transformations are applied:

- **Rotation and Scaling:** Adjusting image orientation and size to account for variations in sketches.
- **Noise Addition:** Simulating imperfections found in hand-drawn forensic sketches.
- **Brightness and Contrast Adjustment:** Standardizing lighting conditions for better recognition.

3.2.2 Normalization and Integration

Pixel intensity values are normalized between 0 and 1 using Min-Max Normalization:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{3.1}$$

This ensures consistency in input values, improving model stability.

3.3 Model Architecture

The system integrates multiple deep learning models for accurate sketch-to-image transformation:

- **CNN (Convolutional Neural Networks):** CNNs are crucial for image processing and feature extraction, making them an essential component of the Sketch-to-Face Recognition project. Built using TensorFlow, the CNN model applies convolutional filters to detect edges, textures, and patterns, gradually learning complex facial structures across multiple layers. This enables effective feature extraction, which is vital for accurate facial recognition.

- **ResNet-50:** ResNet-50 is a deep learning model known for its ability to capture both fundamental and advanced facial features through residual connections. These connections allow deeper networks to be trained efficiently by mitigating the vanishing gradient problem. With 50 layers, ResNet-50 learns hierarchical representations, improving the accuracy of facial recognition by preserving important identity-related characteristics while reducing redundant information.

- **Siamese Neural Network:** A Siamese Neural Network (SNN) is particularly suited for face verification and similarity comparison, leveraging a pair of identical CNNs to process two images independently and generate feature embeddings. By comparing these embeddings using a distance function, such as Euclidean distance, the model determines the similarity between a sketch and a real face, making it highly effective for one-shot learning in facial recognition.

- **AlexNet:** AlexNet, one of the pioneering deep learning models in image classification, remains relevant in face recognition due to its simple yet effective architecture. Consisting of five convolutional layers followed by fully connected layers, AlexNet efficiently extracts meaningful facial features. While it has fewer parameters compared to deeper models like ResNet-50, its lower computational cost makes it a practical choice for scenarios where training speed and efficiency are critical.

3.4 System Flow

This section highlights the flow of the project and includes different diagrams, such as the **Architecture Diagram** and **System Flow Diagram**, to provide a clear understanding of the system's working process. The system is divided into two primary roles: **User** and **Admin**.

3.4.1 Architecture Diagram

The architecture diagram represents the core structure of the forensic sketch-based suspect identification system. It consists of multiple stages, including data preprocessing, model training, and sketch-to-image transformation.

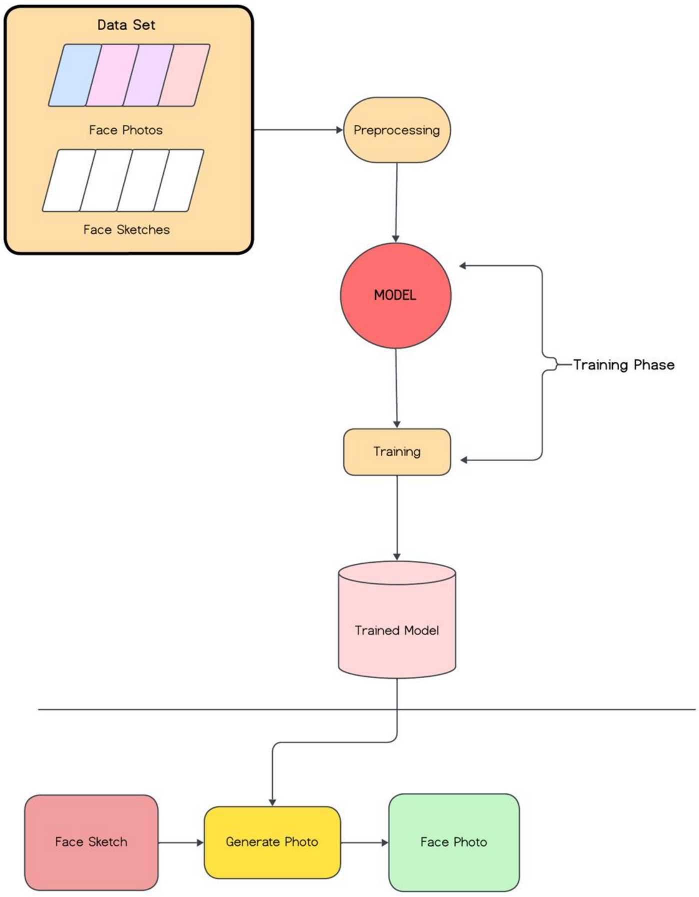

Figure 3.1: Architecture Diagram of the Proposed System

3.4.2 System Flow Diagram

The system flow diagram illustrates the sequence of operations from forensic sketch input to final suspect identification. The process involves dataset preprocessing, model training, sketch transformation, and matching against a criminal database.

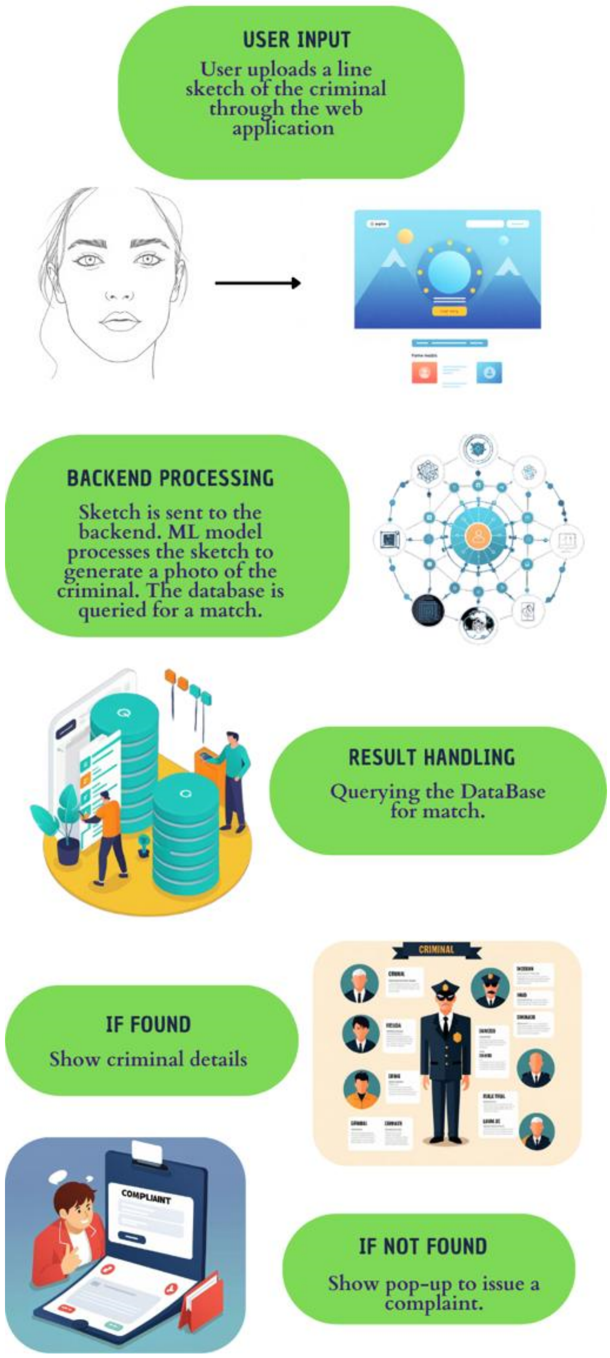

Figure 3.2: System Flow Diagram

3.4.3 User Module

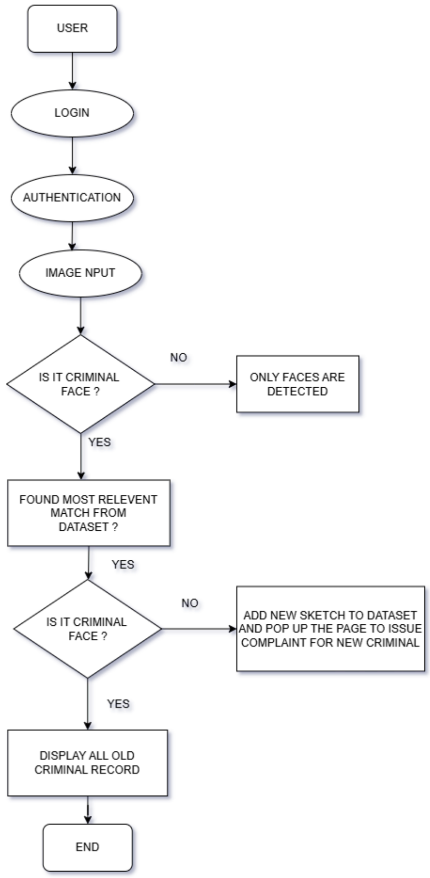

Figure 3.3: Flowchart for User

The user module defines the functionalities available to an end-user (investigator or forensic expert). Users can upload forensic sketches, review generated images, and match them against the criminal database.

- **Upload Sketch:** Users upload a forensic sketch for processing.
- **Image Generation:** The system converts the sketch into a photo-realistic face.
- **Suspect Matching:** The generated image is compared against the criminal database.

- **Report New Suspect:** If no match is found, users can report a new suspect entry.

3.4.4 Admin Module

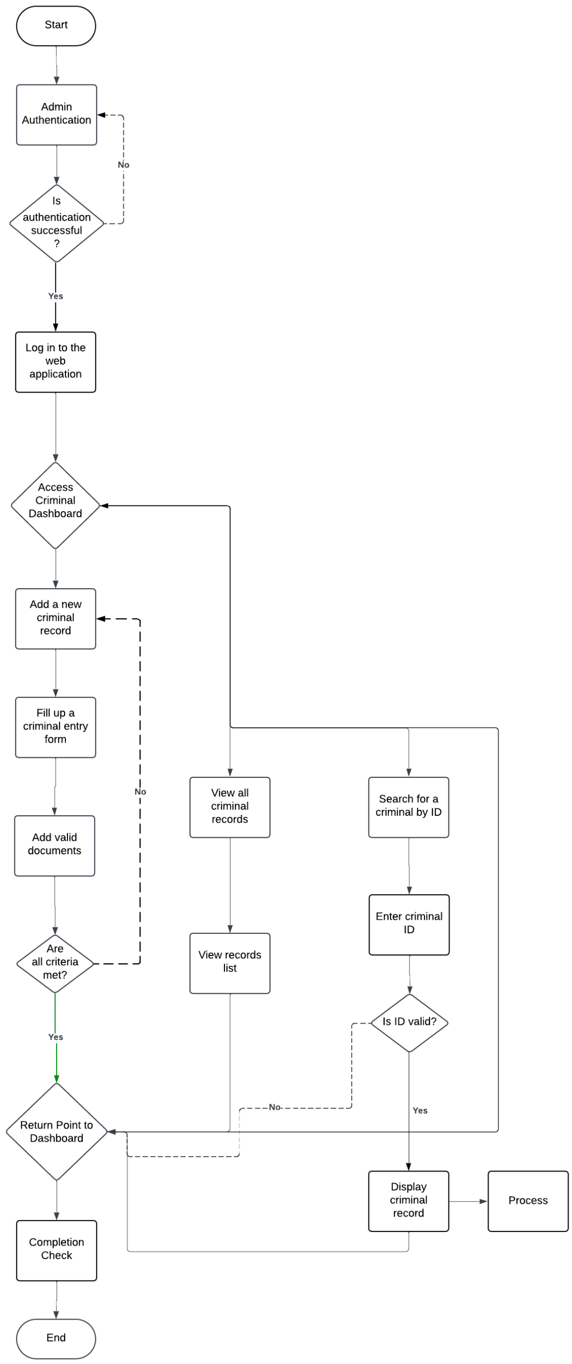

Figure 3.4: Flowchart for Admin

The admin logs into the system for authentication. If successful, they access the **Criminal Dashboard**, which provides three main options:

1. **Add New Criminal:** The admin enters criminal details, uploads necessary documents, and submits the entry. The system ensures all required fields are completed before generating a unique criminal ID.

2. **View All Criminal Records:** The admin can browse and manage all stored criminal records for efficient data retrieval.

3. **Search by Criminal ID:** The admin enters a unique ID to retrieve a specific criminal record. If the ID is invalid, an error message prompts verification.

After completing any task, the admin returns to the dashboard for further actions or logs out.

3.5 Conclusion

The proposed methodology presents an AI-driven system for forensic sketch recognition using deep learning techniques. By integrating CNN, ResNet-50, Siamese Networks, and AlexNet, the model enhances identification accuracy and efficiency. This automated approach reduces manual effort and improves suspect recognition, aiding law enforcement in criminal investigations.

Chapter 4

Results and Discussion

4.1 Results

This section presents the findings obtained from the experimental setup, including the evaluation of machine learning models, accuracy metrics, and visualizations such as heatmaps. The results are reported objectively without interpretation.

4.1.1 Model Performance Evaluation

The accuracy of each model was computed, and results are given in Table 4.1.

Sr. No	**Models**	**Kaggle Dataset**	**Custom Dataset**
1	CNN Tensorflow	56%	69%
2	Siamese CNN	86%	88%
3	ResNet50	70%	75%
4	AlexNet	74%	85%

Table 4.1: Performance comparison of proposed algorithms.

4.1.2 Accuracy Comparison Visualization

To better understand model performance across datasets, a graphical comparison is shown in Figure 4.1.

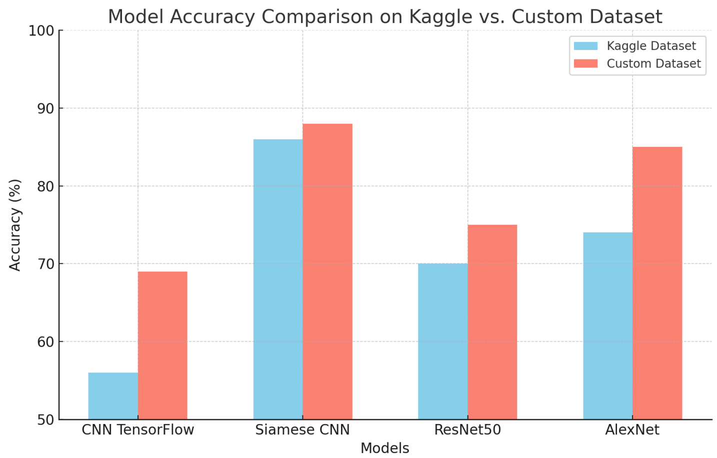

Figure 4.1: Model Accuracy Comparison on Kaggle vs. Custom Dataset

4.1.3 Feature Importance Analysis

The importance of various facial features in CNN-based face recognition was analyzed. Figure 4.2 illustrates the relative importance of different factors affecting model performance.

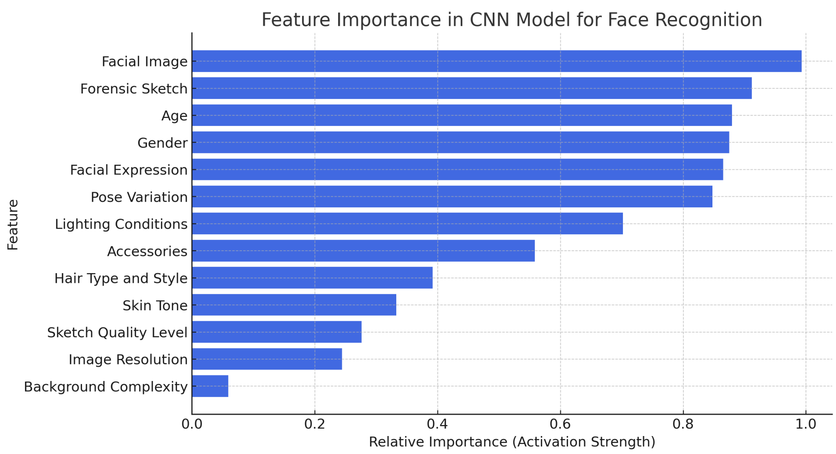

Figure 4.2: Feature Importance in CNN Model for Face Recognition

4.2 Discussion

This section provides an interpretation of the results, comparing findings with prior research and discussing their significance in forensic sketch-to-image transformation.

Overview of Findings

The results indicate that the Siamese Neural Network outperformed other models, achieving the highest accuracy across datasets. This supports the hypothesis that a comparison-based model is highly effective in forensic sketch recognition.

Influencing Factors in Model Performance

Several factors influenced the performance of face recognition models:

- Pose Variations: Frontal facial images resulted in better recognition compared to tilted or side-view images.
- Lighting Conditions: Bright and natural lighting improved accuracy, while dim lighting negatively impacted performance.
- Facial Expressions: Neutral and smiling expressions were easier for the models to recognize.
- Accessories and Appearance: Glasses, hats, masks, and skin tone variations had a noticeable impact on recognition accuracy.

4.2.1 Comparative Analysis with Existing Methods

To understand the effectiveness of our approach, we compare it with traditional and deep learning-based face recognition techniques:

- Traditional Methods (e.g., Eigenfaces, Fisherfaces) struggle with variations in lighting, pose, and sketch quality.
- Classical CNN-based Models such as AlexNet and ResNet50 show moderate performance but are not optimized for forensic sketch matching.
- Siamese CNN, designed for one-shot learning, achieves superior accuracy, demonstrating its suitability for criminal identification applications.

The results confirm that advanced deep learning architectures significantly outperform conventional face recognition methods, particularly in forensic settings.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This study presented the design and development of an AI-driven Sketch-to-Face Criminal Identification System to support forensic investigations. The system employed multiple deep learning architectures, with the Siamese Convolutional Neural Network (CNN) achieving the highest accuracy , followed by AlexNet, demonstrating their practical suitability for sketch-based recognition tasks.

While ResNet50 and CNN with TensorFlow showed moderate performance, their limitations highlighted the importance of model optimization. Factors such as pose variation, lighting inconsistencies, and the presence of accessories were found to significantly impact recognition accuracy.

The study confirms the value of deep learning in forensic sketch analysis, offering a promising solution to aid law enforcement in suspect identification. However, the research also recognizes challenges including limited dataset diversity, computational requirements, and sketch quality inconsistencies that must be addressed for real-world deployment.

5.2 Future Scope

The Sketch-to-Face system holds substantial potential for future expansion and integration into broader criminal justice frameworks. Building on the findings of this study, several key directions can be pursued:

- **Dataset Expansion:** Increasing the size and diversity of forensic sketch datasets to improve generalization across different demographics and artistic styles.
- **Hybrid Model Integration:** Combining various deep learning techniques such as GANs, Siamese Networks, and transformer-based models to enhance system accuracy and robustness.
- **Real-Time Applications:** Developing optimized, lightweight models capable of functioning in real-time environments, including integration with surveillance systems and law enforcement databases.
- **Cross-Domain Compatibility:** Enabling the system to interpret and match sketches created using different artistic techniques and cultural standards.
- **Mobile and Edge Deployment:** Creating versions of the system for mobile and edge devices to support field investigations and remote areas.

With continued advancements in AI and forensic technology, this system has the potential to become a vital tool for national security, intelligence operations, and global law enforcement collaborations.

Chapter 6

References

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Appendix A

Deep Learning Model Details for Sketch-to-Face Recognition

A.1 Model Architecture Details

The table below summarizes the architectures used in the system.

Model	Layers	Feature Extractor	Dataset	Accuracy (%)
CNN TensorFlow	10	Convolutional Layers	Kaggle Dataset	69%
Siamese CNN	8	Pairwise Feature Matching	Custom Dataset	88%
ResNet-50	50	Residual Blocks	Kaggle Dataset	75%
AlexNet	8	Fully Connected	Custom Dataset	85%

Table A.1: Model Architecture Details

A.2 Feature Extraction Timing

Each deep learning model extracts **facial features** at different speeds. The table below provides a comparison.

Model	Feature Extraction Time (ms)	Processing Speed
CNN TensorFlow	10 ms	Moderate
Siamese CNN	7 ms	Fast
ResNet-50	15 ms	Slow
AlexNet	12 ms	Fast

Table A.2: Feature Extraction Timing

A.3 Processor and Computational Requirements

The system was trained on high-performance GPUs. Below are the specifications:

Hardware	**Specification**
Processor	Intel Core i5
RAM	32GB
Dataset Size	100 Images

Table A.3: Processor and Computational Requirements

A.4 Dataset and Preprocessing Details

To ensure high-quality feature extraction, image preprocessing steps were applied:

Step	**Description**
Image Resizing	All images resized to 160x160 pixels
Normalization	Pixel values normalized between 0 and 1
Augmentation	Rotation, Scaling, Brightness Adjustments
Feature Standardization	Using Min-Max Normalization

Table A.4: Dataset Preprocessing Details

A.5 Algorithm Execution Timing

For comparison, we analyzed the execution time of different processes within the system.

Process	**Execution Time (ms)**
Image Preprocessing	5 ms
Feature Extraction (ResNet-50)	15 ms
Matching & Retrieval	8 ms
Displaying Results	3 ms

Table A.5: Algorithm Execution Timing

Appendix B

Fundamentals of Face Recognition and Deep Learning

B.1 Basics of Face Recognition

Face recognition identifies individuals by extracting facial features from images. Key challenges include:

- **Modality Gap:** Differences between sketches and real images.
- **Feature Loss:** Missing fine details in sketches.
- **Pose and Lighting Variations:** Affect accuracy.

B.2 Convolutional Neural Networks (CNNs)

CNNs process images by detecting edges, textures, and patterns through:

- **Convolutional Layers:** Extract key features.
- **Pooling Layers:** Reduce dimensions.
- **Fully Connected Layers:** Perform classification.

B.3 Feature Matching Techniques

Matching a sketch with stored images uses:

- **Euclidean Distance:** Measures similarity.
- **Siamese Networks:** Compare image pairs efficiently.

B.4 Model Training and Optimization

Deep learning models improve recognition through:

- **Loss Functions:** Contrastive loss for similarity learning.
- **Data Augmentation:** Prevents overfitting.
- **Normalization:** Standardizes image values.

B.5 Ethical Considerations

AI-driven face recognition must ensure:

- **Privacy Protection:** Secure biometric data.
- **Fairness:** Avoiding bias in criminal investigations.

Appendix C

Flowchart And Pseudocode

C.1 Combined Flowchart of User and Admin System

The flowchart illustrates the interaction between users and admins in the SketchToFaceRecognition system:

- **User Process:** The user uploads a sketch, which undergoes pre-processing, including resizing and normalization. Features are then extracted using ResNet-50 and compared against the stored portrait database using Euclidean distance.
- **Admin Process:** Admins manage the dataset by adding, updating, and verifying stored portraits. If a match is found, relevant records, such as a criminal background check, are retrieved. If no match is found, the system reports an unknown face.
- The system ensures accuracy by continuously improving recognition capabilities through dataset updates and accuracy computation.
- Admins can also manually review flagged cases to minimize errors and improve recognition efficiency.

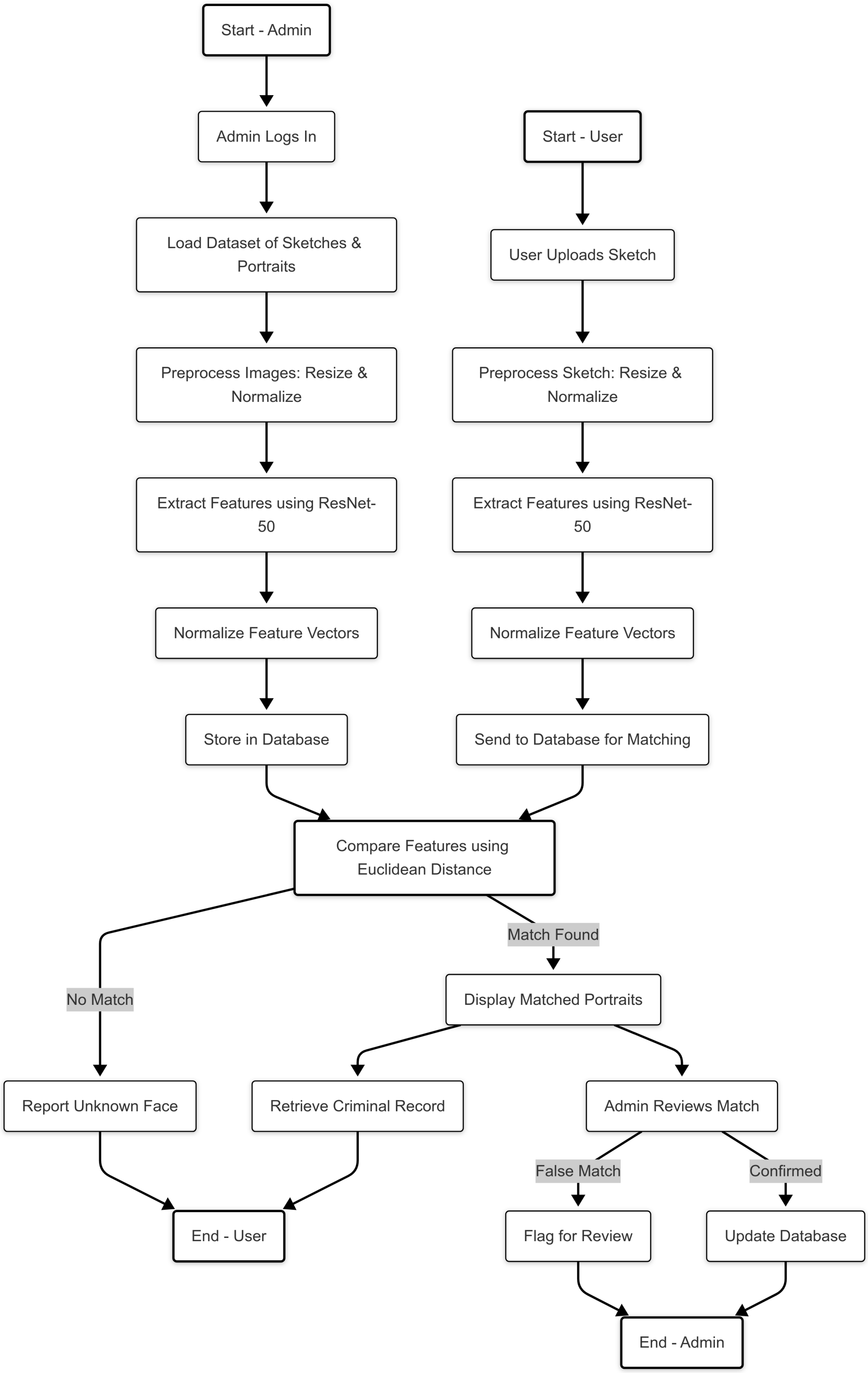

Figure C.1: Combined Flowchart of User And Admin

C.2 Code Snippet for Face Matching

Below is a Python code snippet implementing the face matching algorithm.

```
import cv2
import numpy as np
from tensorflow.keras.applications import ResNet50


model = ResNet50(
    weights='imagenet', include_top=False, pooling='avg'
)
def preprocess_image(image_path):
    image = cv2.imread(image_path)
    image = cv2.resize(image, (160, 160))
    image = image / 255.0
    return np.expand_dims(image, axis=0)


def extract_features(image_path):
    img = preprocess_image(image_path)
    return model.predict(img)


def find_best_matches(sketch_features, portrait_features,
    portrait_filenames, top_k=3):
    distances = np.linalg.norm(sketch_features
    - portrait_features, axis=1)
    sorted_indices = np.argsort(distances)
    best_matches = [portrait_filenames[idx]
```

```
        for idx in sorted_indices[:top_k] ]
    return best_matches
```

This function computes Euclidean distance between a sketch's feature vector and stored portrait vectors, retrieving the closest matches.